Based on the two documents provided, the **GHS_KARAOKE_STUDIO_PLAN.md** is a highly technical implementation blueprint, while **GHS Karaoke.docx** serves as a strategic Product Canvas focused on user experience and the "voice-first" philosophy.

I have merged them into a single, comprehensive **Master Feature Specification**. This document combines the technical code requirements from the Markdown file with the strategic product rules and "Voice-to-Music Intelligence Layer" defined in the Word document.

GHS Karaoke Studio: Master Feature Specification

Voice-First Music Production Pipeline (Unified v1.0)

1. Product Vision & Core Rule

The GHS Karaoke Studio is not a "random AI generator." It is a structured, voice-first production system.

- **The Starting Truth:** The user's recorded voice flow is the foundation. The AI must shape music *around* that flow, never ignoring or replacing the user's musical intent.
- **Target User:** Creative individuals who have musical ideas (melodies, chants, freestyle flows) but lack technical production skills.

2. The Voice-to-Music Intelligence Layer

This layer sits between raw input and final output to ensure the result isn't random. It manages:

- **Flow Analysis:** Detecting phrase timing, pauses, rhythm, and melody contours.
- **Lyric Understanding:** Identifying repeated hooks, possible choruses, and emotional tone.
- **Accompaniment Planning:** Matching the user's flow to a specific "Beat Family" (e.g., Afrobeats, Gospel, or Soft Acoustic).

3. Unified 11-Step Technical Pipeline

This pipeline integrates the technical tools from the Markdown plan with the decision logic from the Product Canvas.

Step	Action	Technology / Tools
1	**Input**	Browser recording, file upload, or WhatsApp voice notes.
2	**Vocal Cleanup**	**Demucs** isolates voice from background noise (generators, street noise).
3	**Flow Analysis**	**Whisper + librosa** extract tempo, key, energy, and timestamps.
4	**Melody Extraction**	**Spotify Basic Pitch** converts humming/singing into MIDI notes.
5	**Lyric Extraction**	**Whisper** provides raw transcription and confidence indicators.
6	**AI Lyrics Assistant**	**Claude** polishes vocabulary, rhymes, and flow while keeping meaning.
7	**Beat Recommendation**	AI suggests styles (e.g., "Mid-tempo Afro-groove") based on flow.
8	**Music Generation**	**Suno/Udio** generates backing tracks using the MIDI melody as a reference.
9	**Vocal Polish**	**RVC** enhances voice quality; Web Audio API adds Autotune/Reverb.

Step	Action	Technology / Tools
10	**Final Merge**	**FFmpeg** combines user voice and AI backing track with balanced levels.
11	**Export**	Standard MP3, Karaoke (timed lyrics), or send to GHS Video Pipeline.

4. Technical Implementation Details

A. Audio Analysis (Python / librosa)

Python

```
def full_audio_analysis(audio_path):
    # Extracts the "Voice Flow Profile"
    y, sr = librosa.load(audio_path)
    tempo, _ = librosa.beat.beat_track(y=y, sr=sr)
    chroma = librosa.feature.chroma_cqt(y=y, sr=sr)
    # Detected Key, BPM, and Energy for Beat Recommendation
    return {"bpm": tempo, "key": keys[chroma.mean(axis=1).argmax()], "energy": librosa.feature.rms(y=y).mean()}
```

B. Lyrics Assistant (LLM Prompting)

Rule: Offer choices (Polish, Simplify, More Pidgin), never overwrite without permission.

- **Syllable Counter:** GHS must warn the user if new lyrics are too long for the recorded melody.

5. Implementation Phases

1. **Phase 1 (MVP):** Voice upload, Cleanup (Demucs), Transcription (Whisper), and Basic Music Build.

2. **Phase 2 (Intelligence):** Advanced flow analysis, chorus/hook detection, and multiple beat alternatives.
3. **Phase 3 (Professional):** Harmony suggestions, ad-lib support, and deep Karaoke timed-lyric tools.

6. Critical "Must-Not" Rules

- **Do not** treat this as one-step random generation.
- **Do not** make the user think like a technical musician.
- **Do not** skip cleanup; background noise ruins AI analysis.
- **Do not** ignore the user's natural vocal rhythm for a "standard" beat.